21-242 Recitation 12

Subspaces — Q4. Images and preimages of subspaces

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

For all subspaces $U_1 \subseteq V$, $U_2 \subseteq W$ and $T : V \to W$, we have that $T^{-1}(U_2) \subseteq V$ and $T(U_1) \subseteq W$ are subspaces.

Course-level solution

Assume T is linear (as is implicit in this section of the course).
Preimage. $T(\vec{0}) = \vec{0} \in U_2$, so $\vec{0} \in T^{-1}(U_2)$. If $\vec{v}, \vec{w} \in T^{-1}(U_2)$ and $a, b \in \mathbb{F}$, then $T(\vec{v}), T(\vec{w}) \in U_2$, hence $T(a\vec{v} + b\vec{w}) = aT(\vec{v}) + bT(\vec{w}) \in U_2$ because U_2 is a subspace. Thus $a\vec{v} + b\vec{w} \in T^{-1}(U_2)$.
Image. $T(\vec{0}) = \vec{0} \in T(U_1)$. If $T(\vec{v}), T(\vec{w})$ are in $T(U_1)$ with $\vec{v}, \vec{w} \in U_1$, then $a\vec{v} + b\vec{w} \in U_1$, so $aT(\vec{v}) + bT(\vec{w}) = T(a\vec{v} + b\vec{w}) \in T(U_1)$.

Later-course solution (using ideas from the rest of the handout)

These are the statements $\ker T = T^{-1}(\{\vec{0}\})$ and $\operatorname{im} T = T(V)$ as special cases. After rank-nullity one also knows dimensions: $\dim T(U_1) = \dim U_1 - \dim(U_1 \cap \ker T)$ and $\dim T^{-1}(U_2) = \dim(U_2 \cap \operatorname{im} T) + \dim \ker T$.